Politecnico di Torino

Qubit Electronics
Lab 3

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1 Introduction

This laboratory focuses on the analysis of a double quantum dot system, in which quantum confinement results in distinct energy levels, differentiating from the continuous electronic bands typical of bulk materials. Throughout this exercise, you will develop skills in examining such structures through the application of Poisson and Schrödinger equations.

In this laboratory, you will use two software: GMSH and QTCAD. Gmsh is an open-source 3D finite element mesh generator with a built-in CAD engine and post-processor. The documentation is available following this link: Gmsh Documentation. QTCAD® (Quantum-Technology Computer-Aided-Design) is a finite-element (FEM) simulator used to predict the performance of spin-qubit devices before their production. The documentation of this tool is available here.

2 Mesh generation

1. Download the zip file `Lab3-TutorialQTCAD.zip`.
2. Unzip the downloaded file on your local machine or server. If you are using only the prompt interface, you can upload the folder using FileZilla or any other suitable file transfer method.
3. Once unzipped, navigate to the extracted folder. You will find the following structure:
   - A Python script file named `double_dot_stability.py`, which is the main script to run the simulation.
   - Two folders named `meshes` and `output`.
     - The `meshes` folder contains a file with the extension ".geo", which represents the mesh to be compiled for the simulation.
     - The `output` folder is currently empty but will store the output files generated during the simulation.
4. Then, open the mesh file `dqdfdsoi.geo` in the folder meshes and review each line of the code.

Question: Explain the process used in the code to initialize the structure and how the dimensions and areas are defined.

Question: Explain how the mesh structures are defined in the code and how the rectangles and various domains of the structure are defined. To better understand the definition of rectangles within the mesh, refer to the tool manual.

Question: Observing how the different surfaces are defined in the code, explain how one could define a solid.

At the end of the code, there are a few lines where the simulation results are saved in two different formats.

Question: Explain the difference between the two file formats and why we choose to obtain them.

Open the terminal in the main folder (the unzipped archive) and write the following line to activate the conda environment for QTCAD in case you are using the laboratory PC

```
conda activate D:\.conda\envs\qtcad
```

Otherwise (using SSH or X2GO), it is just necessary to write:

```
conda activate qtcad
```

Then, to compile the mesh, enter the following line of code in the terminal.

```
gmsh dqdfdsoi.geo
```

Question: Provide a screenshot of the three-dimensional structure in the clearest manner possible.

After visualizing the mesh, look into the folder containing the file and specify what type of files have been automatically created.

Open the `geo_unrolled` file in the folder meshes, which appears after the execution of the gmsh command, to understand the contents.

Question: What are the dimensions of the whole structure along the x, y, and z coordinates? In addition, try to comprehend the base classes and how they are declared. Refer to the manual for further information.

3 Schrodinger and Poisson: understanding the script file

Open the file `double_dot_stability.py`

Look at these two lines of code:

```
scaling = 1e-9
mesh = Mesh(scaling, str(path_mesh / "dqdfdsoi.msh2"))
```

Question: What is the function of these two lines of code, what are the parameters inserted for the Mesh class?

Subsequently, look at these two lines of code:

```
dvc = Device(mesh, conf_carriers='e')
dvc.set_temperature(0.1)
```

Question: Explain the function of these lines of code defining what the arguments passed are. Then, explain the line of code related to the temperature, indicating the operating temperature with the appropriate unit of measurement.

Continue by inspecting the code meticulously and the designated region names.

Question: Identify the materials assigned to each region and compare the names of script files with the defined names in the mesh.

By looking at the subsequent lines of code, we can see lines of code like `new_gate_bnd`.

Question: Explain the code's arguments, discern within the code the regions of the dot, and the corresponding materials that compound it.

Question: Continuing to observe the same code `double_dot_stability.py`, understand when the class is defined, when the methods for solving Poisson and Schrodinger equations are instantiated, and when they are solved. Report the lines.

4 Simulation: Poisson and Schrodinger

Once the Conda environment QTCad is activated, you need to enter the Python command line

```
python double_dot_stability.py
```

The results will be in the "Output" folder.
Open Paraview, click with the right button of the mouse on the left window, and select "Open" to read subsequent files.

4.1 File EC.vtu

The next step is to open the Paraview program, and import the file ending with EC.vtu. Subsequently, open the panel on the left, click "Apply" in the bottom menu, then "Slice" in the top menu, and set it with Z-normal at a height equal to -2.5. Finally, in the section "coloring" select "Rescale to custom range", and set the range in the interval $[0, 0.1]$.

Question: Why do we choose to slice at this height and not, for example, at -11? After obtaining the output image, explain what is shown after the slice command, display the image, and describe it briefly also explaining which is the referring color that characterizes the zones that contain a double quantum dot.

The next step is to click on "Plot Overline", then in line parameters: vary the variable y all over the domain, and set x values to 0 and z remains at -2.5.

Question: Describe the plot, show how the potential changes in the domain and what the minima represent; and explain if the behavior in the central region is similar to the expected one or not.

4.2 File N.vtu

Subsequently, open the file ending with N.vtu, in the section "Coloring" select "n", and click on "Slice" with z-normal at a height equal to -2.5, exactly as with the previous file. Finally, press "Coloring" and then "Rescale to data range".

Question: Observing the simulation just obtained, lower the Set Range value, for example, to 1e18. Knowing that this represents a simulation obtained as a result of Poisson's solver, explain why the central area of the output image, where the dots are located, appears to be close to zero.

4.3 File Q.vtu

Finally, open the file ending with Q.vtu, "Slice" with Z-normal at the same height as before, and observe the result both for the ground state and for the first excited state.

Question: Explain what the displayed file represents: what can you notice in the plot?

We have to remember that the mesh introduces asymmetries that give rise to a non-symmetric potential and the localization of a single quantum dot. Moreover, employing an adaptive mesh may result in variations of this process across simulations due to the inherent variability in mesh configurations.

Additionally, press "Plot Overline" selecting y at the midpoint.

Question: Explain how the curve behaves in both the ground state and the first excited state, compare the two curves, and indicate their respective configurations.